7. ESTIMATION OF DEOXYRIBO NUCLEIC ACID(DNA)

(Demonstration only)

Aim:

To quantify the deoxyribonucleic acid content in the given test solution.

Principle:

The deoxyribose moiety of the DNA forms a hydroxyl levulaldehyde in TCA solution. This reacts with diphenylamine to give a blue colour which can be read at 565 nm.

Reagents:

1. **Dische's reagent:** Dissolve 1.5g of crystalline diphenylamine in 100 ml of glacial acetic acid and add 1.5 ml of conc. sulphuric acid.The reagent is stored in dark bottles. At the time of use add 0.1 ml of aqueous acetaldehyde (16mg/ml) for each 20 ml of the reagent.
2. 1N perchloric acid:
3. 5mMSodium hydroxide:
4. **Standard DNA solution:** A stock solution is prepared by dissolving 100mg of DNA in 5M sodium hydroxide. Working standard solution is prepared by mixing 10ml of the stock standard with 1N perchloric acid and heating at 700C for 15 minutes.

Procedure:

0.5-2.5ml of working standard DNA solution is pipetted out into five different test tubes and labelled as S1-S5 with the concentration of 50-250µg. 1.0ml of the test solution is pipetted out into a test tube and labelled as T_1. Final volume is made upto 3.0ml using distilled water in all the tubes. A blank is also prepared simultaneously comprising of 3.0ml of distilled water. 2.0ml of diphenyl amine (Dische's reagent) is added to all the test tubes. The initial turbidity formed disappears gradually. The tubes are covered with glass marbles and heated in a boiling water bath for 10-15 min. The test tubes are then cooled under tap water and the blue colour developed is measured at 565nm in a spectrophotometer.The control or blank contains only water and the Dische's reagent.

A standard graph is drawn with optical density in Y- axis vs. concentration of DNA in X axis. The amount of DNA present in the given test solution is then calculated.

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